

THIRANEX INTERNSHIP ACTIVITY

TITLE:

**“IPL DATA CLEANING AND VISUALIZATION
PROJECT”**

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ABSTRACT

The data is one of the most valuable assets that has come along in the modern world and organizations have turned to data for insights to guide their decisions. Raw data from a real-world source, however, tends to be incomplete, inconsistent and confusing when presented in its raw state to the naked eye. Thus, data cleaning and data visualization are integral parts of making sense of raw data and making it actionable. Data cleaning can help identify and correct errors, missing data, and duplicate records, and data visualization can help make complex data easier to understand and analyse.

The project aims to analyse and present historical Indian Premier League (IPL) match data, focusing on cleaning and visualizing the information using data analytics tools in Python. The IPL is a popular and competitive cricket tournament that generates a vast amount of data on teams, stadiums, tosses, game results, and season performance. This information can be used for exploratory data analysis and gain interesting insights into team performances and match characteristics.

The data used for this project includes the IPL match history data such as the season, city, teams playing, toss winner, toss decision, match winner, venue and winning margin. Data was analysed to find any missing, inconsistent and duplicate records. The dataset was then pre-processed using appropriate techniques for cleaning and improving the quality of the data. Once the data was cleaned, different visualizations were created using Matplotlib and Seaborn libraries, to discuss various aspects of the IPL data. The visualizations comprise of match analysis by the seasons, winning team distribution, pattern of toss decisions, venue analysis, city-wise match distribution, winning margins, and correlation analysis.

The output visualizations provide insights into key trends and patterns in IPL's history. The project reveals the potential of data cleaning and visualization methods in sports data to discover valuable information and facilitate analytical decision-making. This book offers hands-on experience with data preprocessing, exploratory data analysis, and storytelling through visualization in Python, and introduces the reader to the significance of data analytics in sports and business applications.

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1. INTRODUCTION

Large quantities of data are created every day from many sources in today's data-driven world. But raw data is generally incomplete, inconsistent, or hard-to-understand in raw format. Hence, data cleaning and data visualization is a crucial element in converting raw data to useful information for analysis and decision making. Data cleaning can be used to improve the quality of data by cleaning up data with missing values, removing duplicate data, and ensuring consistency; Data visualization can be used to visually present data, making it easier to understand and analyse

One of the emerging applications of data analysis is sports analytics, which involves analyzing historical sport-related data to find patterns, assess performance, and derive insights. As cricket is one of the most popular sports in the world, it generates a vast amount of data which can be analysed to gain insight into many aspects of the sport. The Indian Premier League (IPL) is one of the most successful T20 cricket tournaments and spits out huge data pertaining to teams, venues, toss decisions, match outcome, and seasonal performance.

The aim of this project is to use Pandas, Matplotlib, and Seaborn libraries in Python to clean and visualize IPL match data. All the data in the dataset was examined to find missing and inconsistent data and then apply to appropriate preprocessing methods to enhance data quality. Patterns for team performance, toss decisions, match venues, margins of victory and even match distribution by season were then explored through the creation of various visualizations.

2. DATASET DESCRIPTION

- **Dataset Name**
IPL Data Set
- **File Used**
matches.csv
- **Source**
Kaggle IPL Dataset

Match No.	Match ID	Season	Venue	Date	Team 1	Team 2	Team 1 Score	Team 2 Score	Match Type	Match Status	Winning Team
1	2	IPL-2017	Pune	06-04-2017	Mumbai Indians	Rising Pune Supergiant	190	181	field	normal	Rising Pune Supergiant
2	3	IPL-2017	Rajkot	07-04-2017	Gujarat Lions	Kolkata Knight Riders	176	176	field	normal	Kolkata Knight Riders
3	4	IPL-2017	Indore	08-04-2017	Rising Pune Supergiant	Kings XI Punjab	190	181	field	normal	Kings XI Punjab
4	5	IPL-2017	Bangalore	08-04-2017	Royal Challengers Bangalore	Delhi Daredevils	190	181	bat	normal	Royal Challengers Bangalore
5	6	IPL-2017	Hyderabad	09-04-2017	Gujarat Lions	Sunrisers Hyderabad	190	181	field	normal	Sunrisers Hyderabad
6	7	IPL-2017	Mumbai	09-04-2017	Kolkata Knight Riders	Mumbai Indians	190	181	field	normal	Mumbai Indians
7	8	IPL-2017	Indore	10-04-2017	Royal Challengers Bangalore	Kings XI Punjab	190	181	bat	normal	Kings XI Punjab
8	9	IPL-2017	Pune	11-04-2017	Delhi Daredevils	Rising Pune Supergiant	190	181	field	normal	Delhi Daredevils
9	10	IPL-2017	Mumbai	12-04-2017	Sunrisers Hyderabad	Mumbai Indians	190	181	field	normal	Mumbai Indians
10	11	IPL-2017	Kolkata	13-04-2017	Kings XI Punjab	Kolkata Knight Riders	190	181	field	normal	Kolkata Knight Riders
11	12	IPL-2017	Bangalore	14-04-2017	Royal Challengers Bangalore	Mumbai Indians	190	181	field	normal	Mumbai Indians
12	13	IPL-2017	Rajkot	14-04-2017	Rising Pune Supergiant	Gujarat Lions	190	181	field	normal	Gujarat Lions

Fig 1. Dataset Sample

Features Used

Feature	Description
Season	IPL season
City	Match location
Team1	First team
Team2	Second team
Toss Winner	Team winning the toss
Toss Decision	Bat or Field
Winner	Match winner
Win By Runs	Winning margin by runs
Win By Wickets	Winning margin by wickets
Venue	Match venue

3. DATASET CLEANING PROCESS

Data cleaning was performed to improve dataset quality before visualization.

Steps Performed

1. Loaded dataset using Pandas.
2. Checked dataset structure and information.
3. Identified missing values.
4. Removed missing records.
5. Checked for duplicate entries.
6. Removed duplicate records.
7. Verified cleaned dataset dimensions.

Missing Value Analysis

```
print(df.isnull().sum())
```

id	0
Season	0
city	7
date	0
team1	0
team2	0
toss_winner	0
toss_decision	0
result	0
d1_applied	0
winner	4
win_by_runs	0
win_by_wickets	0
player_of_match	4
venue	0
umpire1	2
umpire2	2
umpire3	637
dtype:	int64

Fig 2. Missing Value Analysis

Cleaned Dataset Shape:

Displays the final number of rows and columns after cleaning.

```
print(df.shape)
```

```
(118, 18)
```

Fig 3. Cleaned Dataset Shape

4. DATA VISUALIZATION AND ANALYSIS

4.1. Visualization: Matches Played Per Season

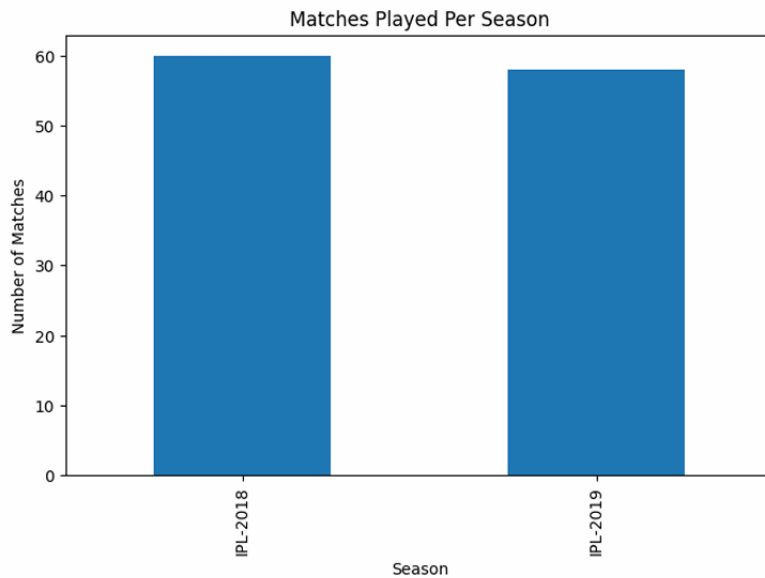


Fig 4. Bar Chart for Matches Played Per Season

4.2. Visualization: Top Winning IPL Teams

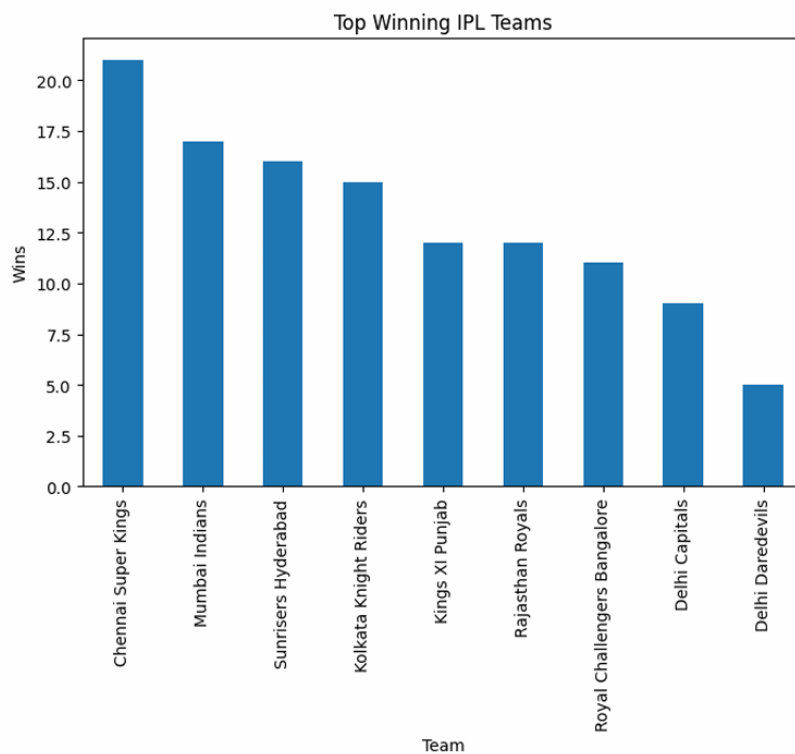


Fig 5. Bar chart for Most Winning Teams

4.3. Visualization: Toss Decision Analysis

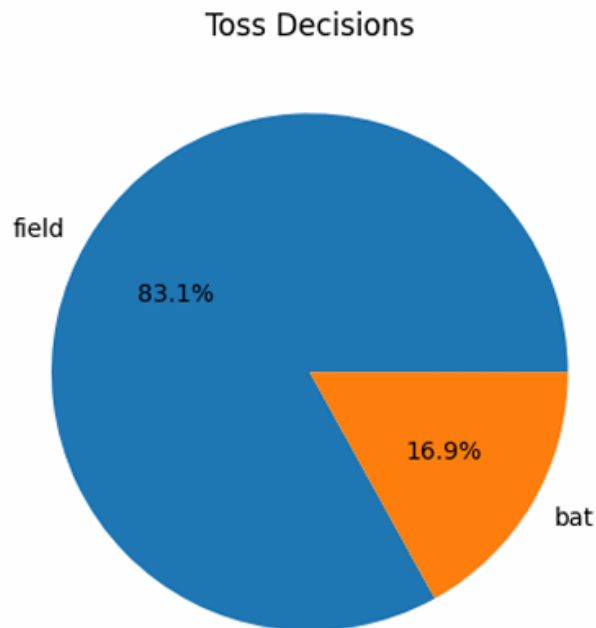


Fig 6. Toss Decisions Pie Chart

4.4. Visualization: Winning Margins by Runs

Code:

```
import matplotlib.pyplot as plt

# Remove matches won by wickets (win_by_runs = 0)
runs_margin = df[df['win_by_runs'] > 0]

# Top 10 most common winning margins
top_runs = runs_margin['win_by_runs'].value_counts().head(10)

plt.figure(figsize=(10,6))
top_runs.plot(kind='bar')

plt.title("Top 10 Most Common Winning Margins by Runs in IPL")
plt.xlabel("Winning Margin (Runs)")
plt.ylabel("Number of Matches")
plt.xticks(rotation=0)

plt.tight_layout()
plt.show()
```

Bar Chart

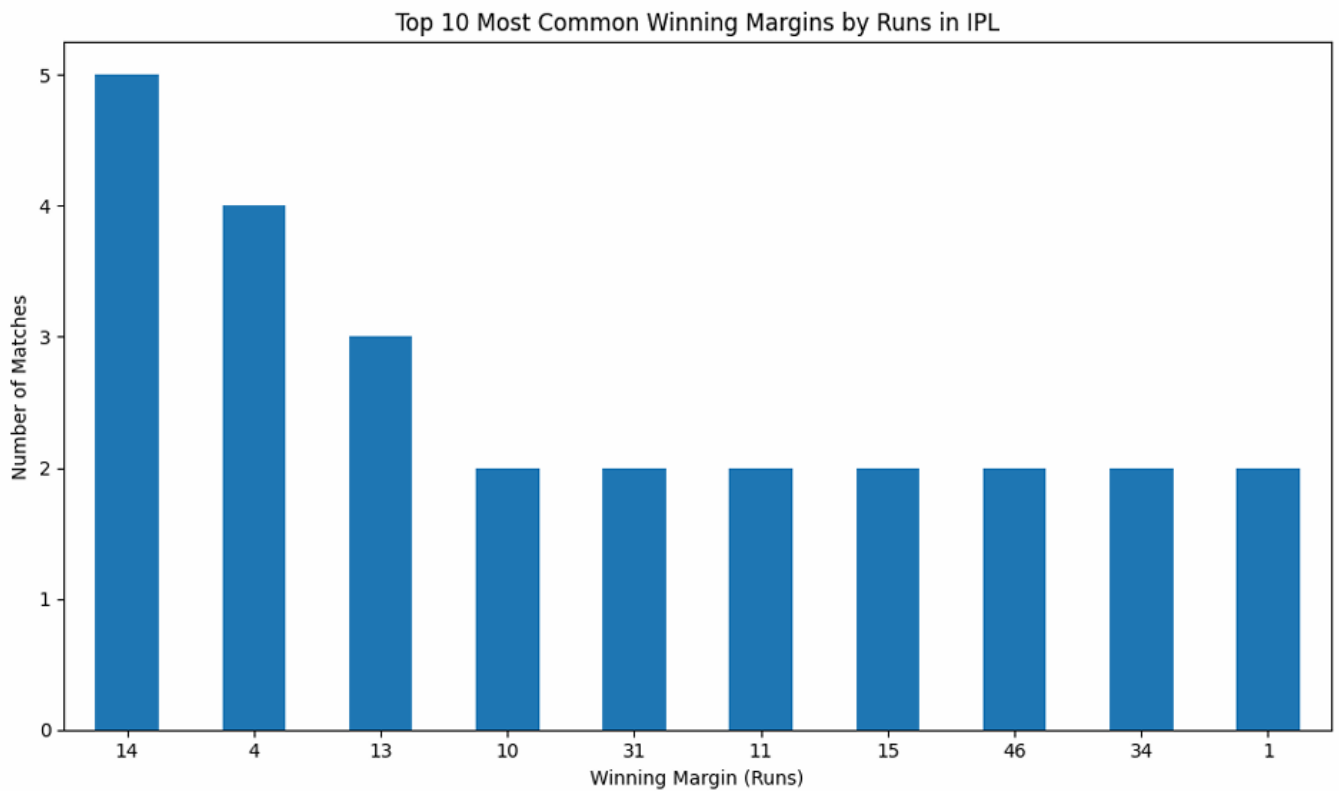


Fig 7. Winning margin by Runs Bar Chart

4.5. Visualization: Winning Margins by Wickets

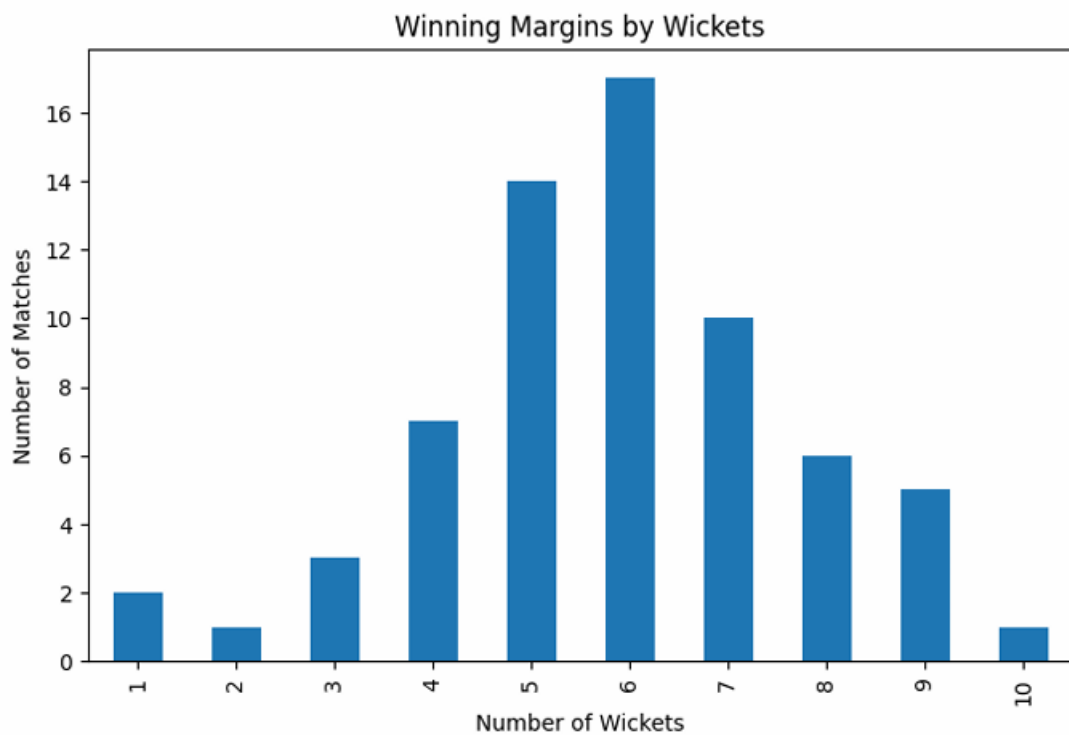


Fig 8. Win Margins by Wickets Bar Chart

4.6. Visualization: Top IPL Venues

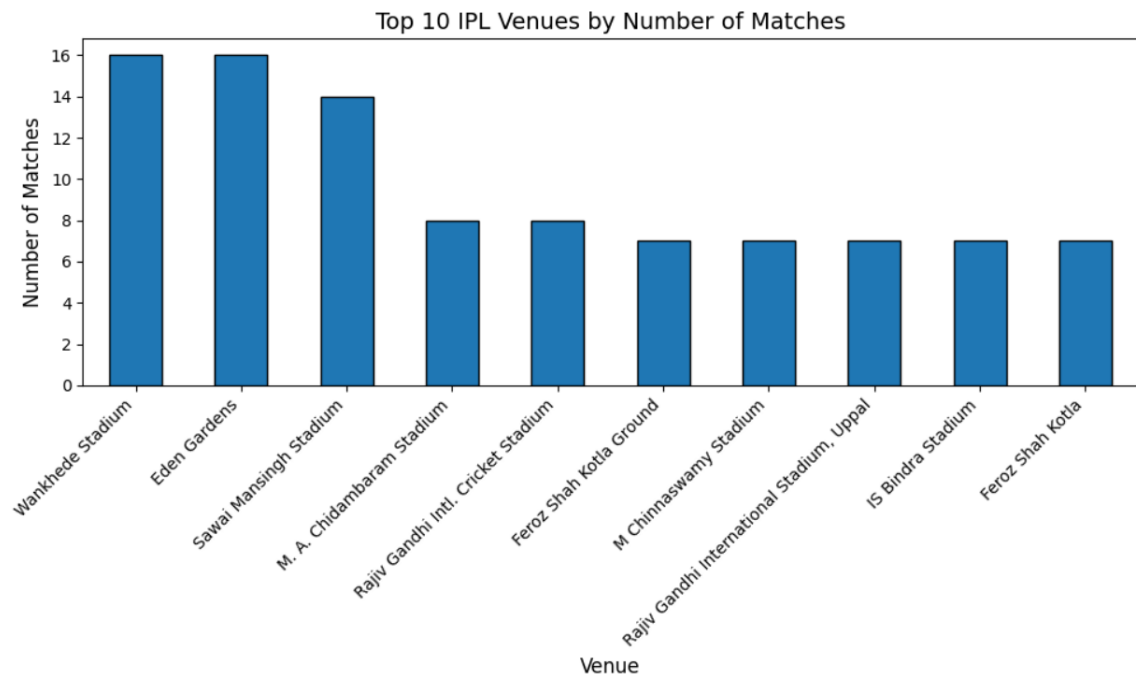


Fig 9. Top IPL Venues Chart

5. CONCLUSION

The IPL Data Cleaning and Visualization Project effectively showcased how raw cricket data can be processed and analysed to derive meaningful insights. The IPL dataset was thoroughly analyzed for missing values, duplicate data, inconsistencies, and corrected accordingly with suitable data cleaning techniques. This made sure that the data set was accurate, reliable and appropriate for analysis.

Following pre-processing, several visualizations were developed to investigate various facets of the IPL matches such as Season-wise match distribution, Team performance, toss decision, venue popularity, winning margin and the distribution of winners. These visualisations were useful in identifying key trends and patterns in historical IPL data. The analysis provided insights, including which teams dominated the competition, which teams the players preferred to bat first after the toss, which venues were commonly used and what were the typical winning margins in IPL matches.

The project emphasizes the need for data cleansing and data visualization, both of which are key parts of the data analytics journey. Complex information is made easier to understand and interpret by being presented in an informative, graphical format. In addition, this project offered hands-on experience with Python, Pandas, Matplotlib, and Seaborn to conduct exploratory data analysis and visual storytelling.

The project overall showcases the ability of data analytics methods to be applied successfully to sporting data to gain value, assist decision making, and help understand past trends. Enhanced statistical analysis, interactive dashboard features, and predictive modeling could be part of future work to further improve data analytics for the IPL and provide deeper insights.